

City Tree Equity Analyzer

STANDARDIZING ENVIRONMENTAL METRICS ACROSS EUROPEAN URBAN LANDSCAPES

The Crisis of Urban Canopy

SPENCER ANDERSON

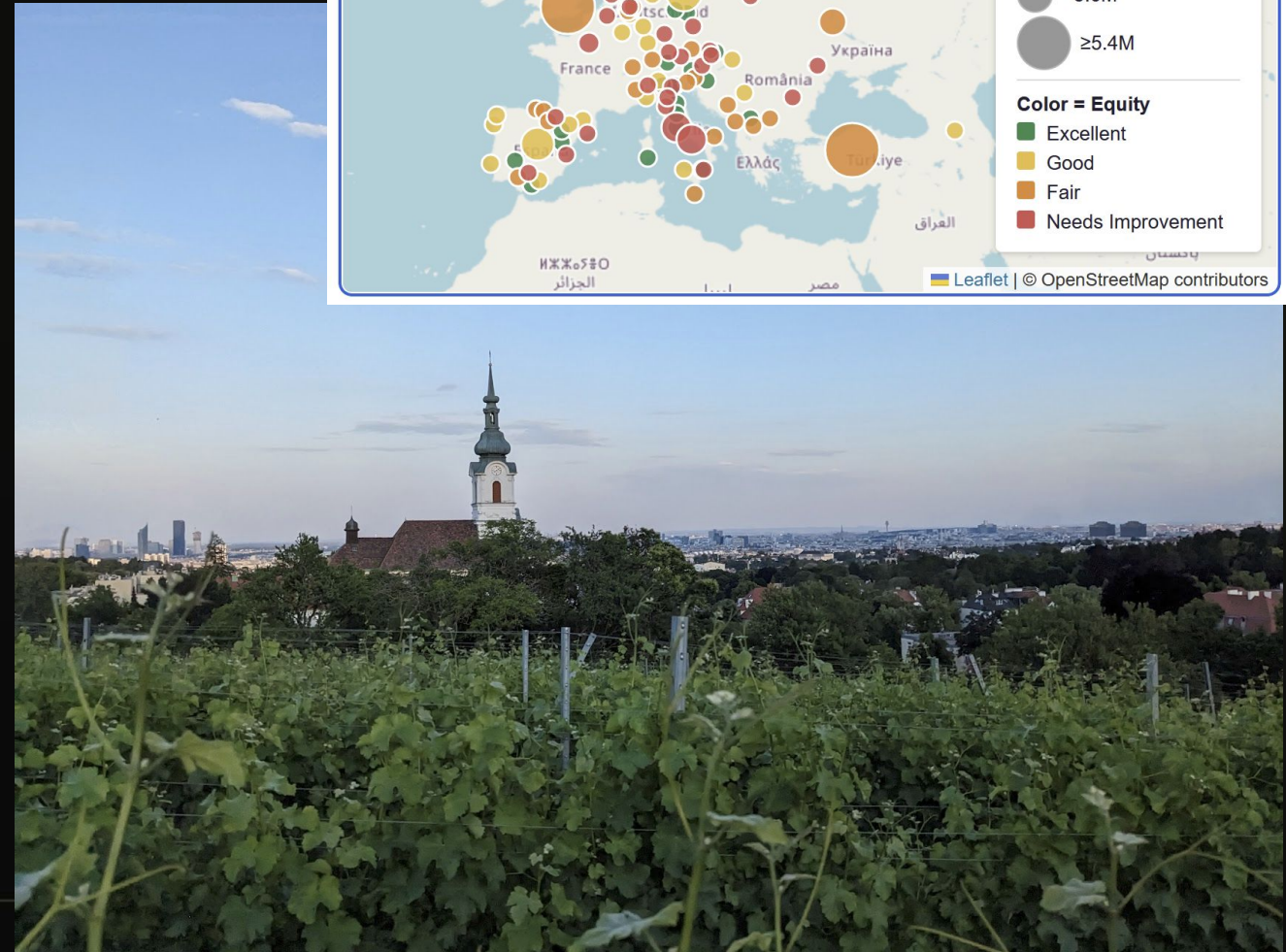
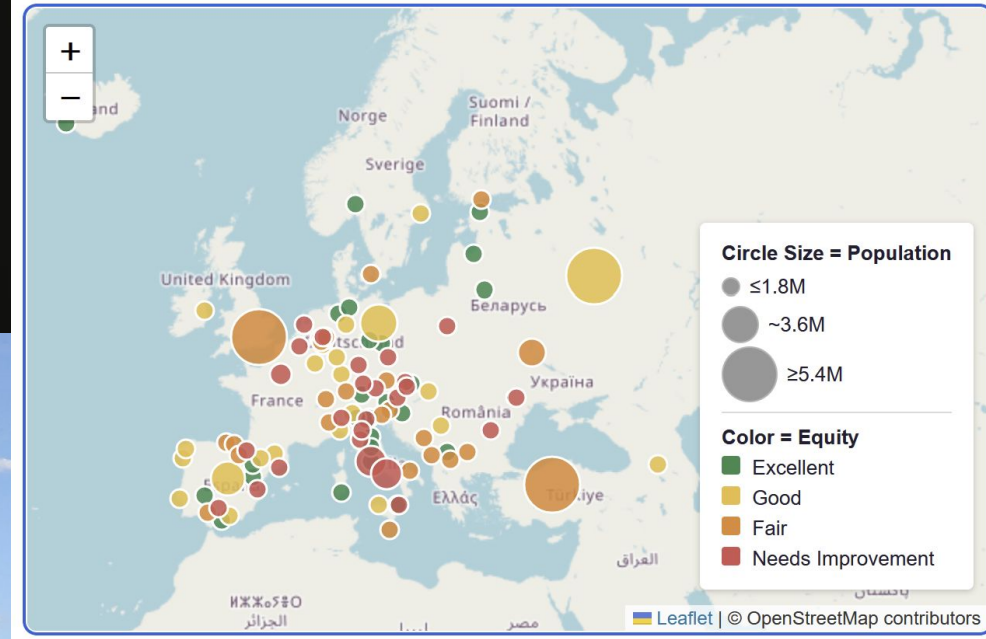
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Urban Inequity & Access

Tree distribution in European urban centers exhibits significant structural imbalance. While certain districts provide ample canopy, high-density metropolitan cores often leave residents with minimal green infrastructure.

The data reveals that presence of trees does not equate to equitable access. Metrics must weight tree canopy against population pressure to identify the true "Green Gap."

City Map — Geographic Distribution of Tree Equity

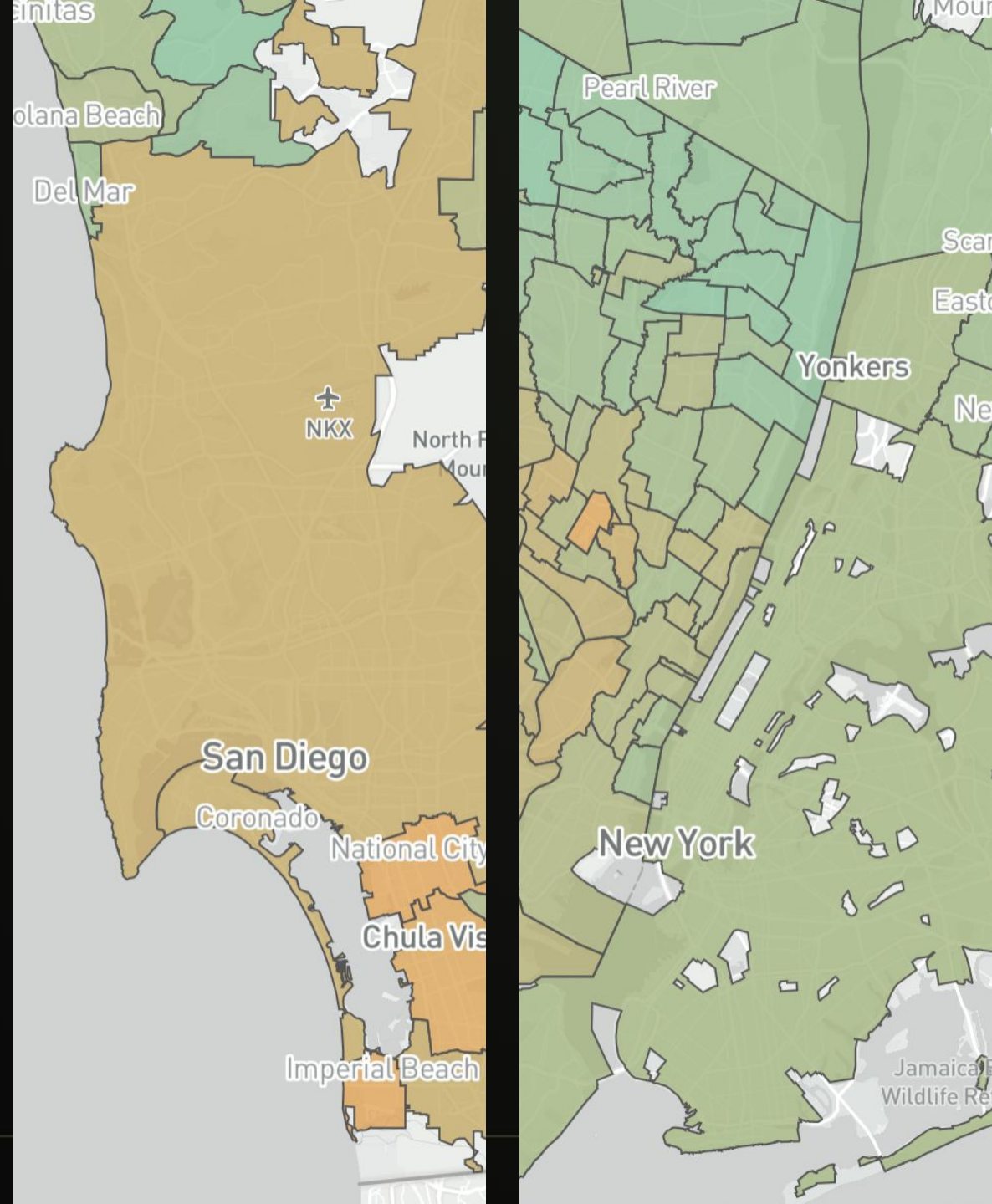


Transatlantic Contrast

US vs Europe Methodology

United States: Services such as TreeEquityScore.org evaluate cities at the micro-neighborhood level, leveraging unified Census data for sociodemographic weighting.

Europe: The absence of standardized cross-border neighborhood data necessitates a macro-level benchmark. The Analyzer fills this gap using FUAs to establish the first international equity comparison across 28 nations.



The Data Foundation

Source	Description	Records
Eurostat TGS00096	NUTS population data (2014–2024)	227 regions
Copernicus Urban Atlas 2021	FUA boundaries in GeoPackage format	768 metropolitan areas
Copernicus TCD 2015	Tree Cover Density raster at 20m resolution	1 raster covering EU
NUTS2021	NUTS classification with metropolitan labels	229 entries
Wikipedia	European country and city data	51 countries, 30 cities

The Processing Pipeline



Ingest

Synthesis of disparate demographic and regional datasets into a unified SQLite relational structure for high-performance querying.



Extract

Execution of precision spatial clips on 1GB+ tree rasters using city-specific Functional Urban Area boundary geometries.

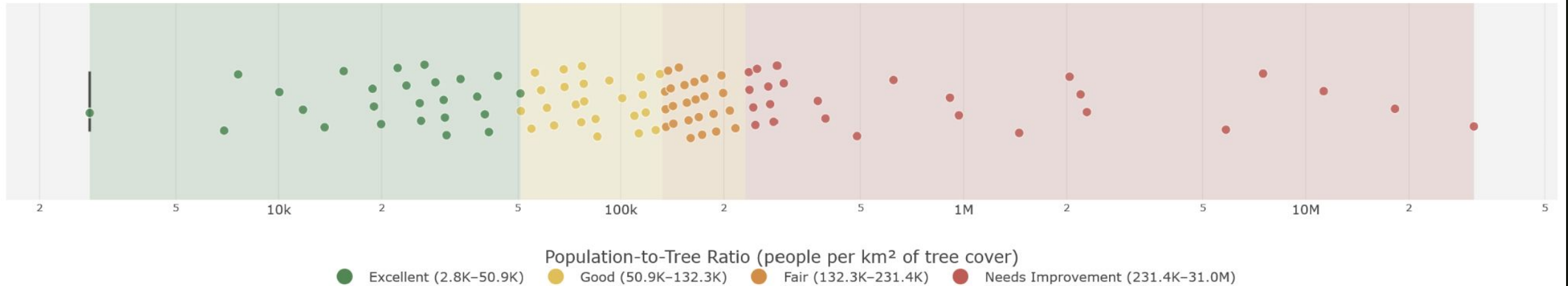


Normalize

Development of cross-border equity scores by integrating population density with high-resolution canopy surface area metrics.

Categorizing Equity via Quartiles

City Tree Canopy Equity Distribution by Quartile



Excellent Access

Defined as the top quartile of the distribution.. These cities exhibit minimal population pressure on existing urban green infrastructure.

Needs Improvement

Defined as the lowest quartile of the distribution. High density severely outpaces canopy availability, signaling priority investment zones.

Methodology Issues

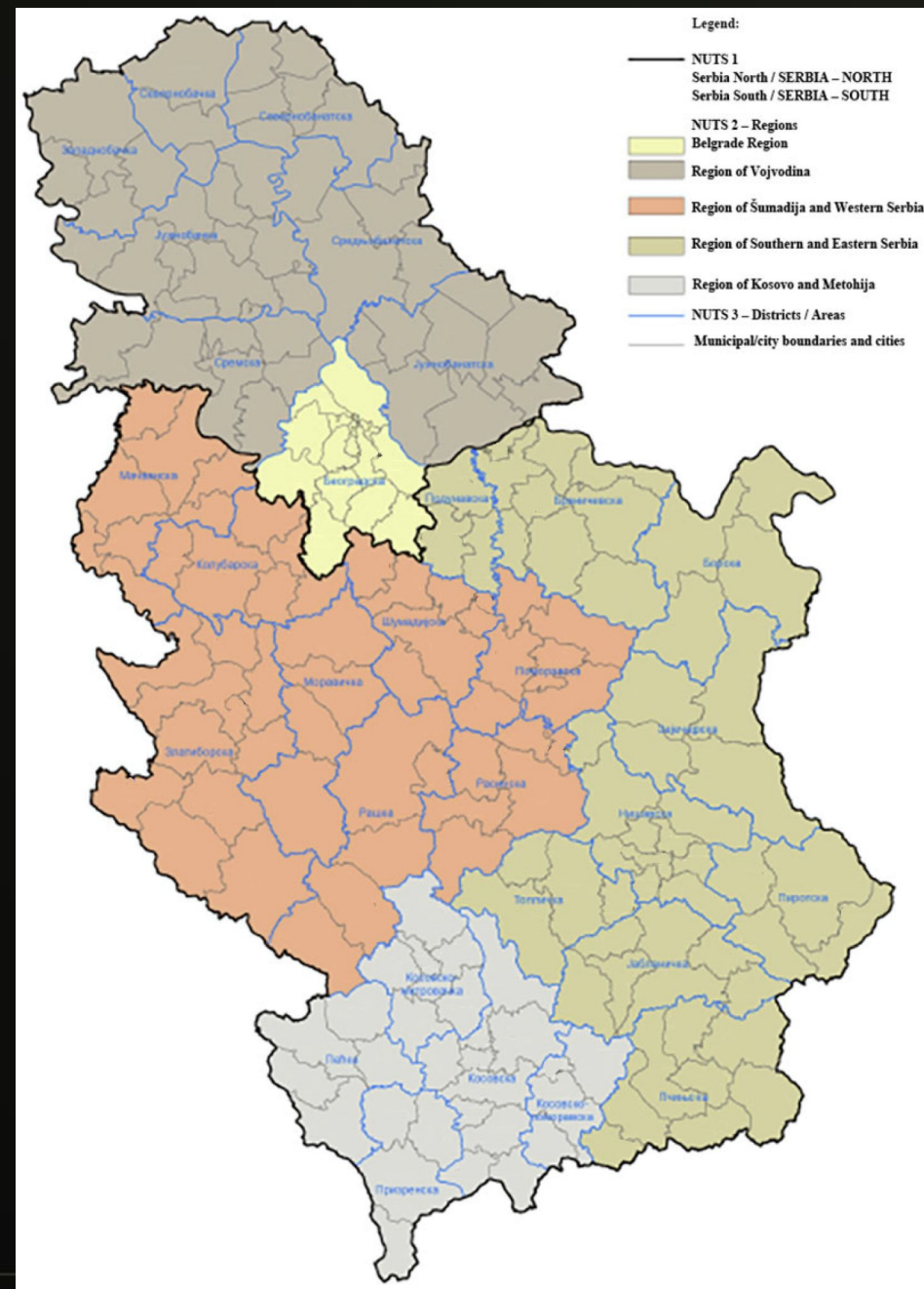
Difference in reporting area

NUTS2 (Administrative): Used for Eurostat population data. These political lines often encompass vast non-urban areas, distorting density metrics.

FUA (Functional): Used for Copernicus tree data. These represent the actual economic and commuting footprint of a city center.

Disparity Solution: The pipeline clips tree rasters strictly to the FUA (the living reality) and maps population density from corresponding NUTS2 territories to normalize results.

The primary hurdle in calculating ***Equity Status = Population / km² of Tree Cover*** is that a NUTS2 administrative polygon and an FUA economic polygon rarely overlap perfectly.



Questions?

CITY TREE EQUITY ANALYZER | PROJECT SUMMARY

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